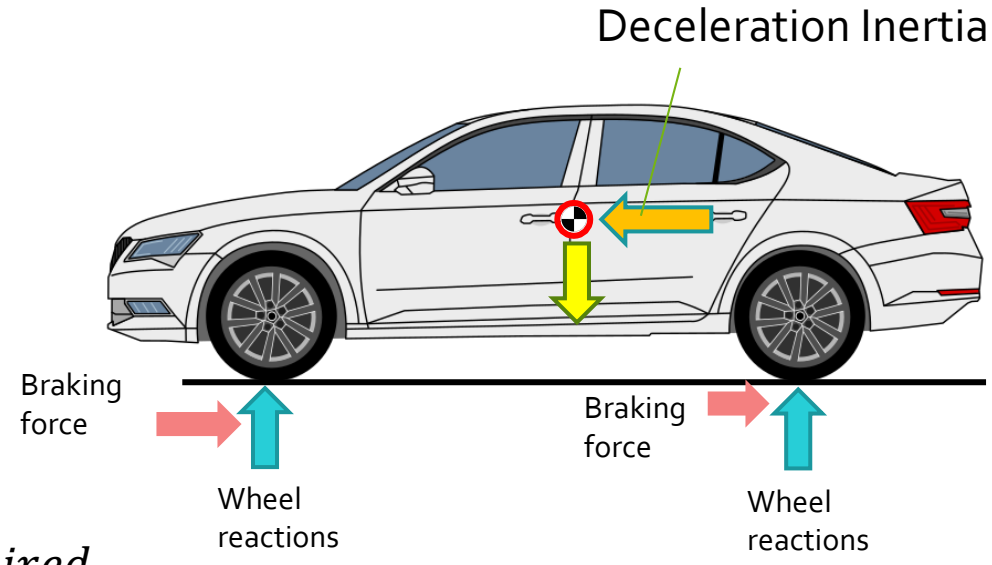
Braking Calculations Problem

A vehicle with a mass 2500 kg is moving at 50 kmph and is coming to halt in 3 seconds.
What is the Total Braking force required at the wheels ?



Deceleration in m/s^2 $\frac{0 - 13.889}{3} = -4.629\ ms^{-2}$

$$Braking\ force\ required\ = mass\ * deceleration\ required$$

$$Braking\ force\ required\ = 2500\ * 4.629$$

$$Total\ Braking\ force\ required\ = 11572.5\ N$$

$$Braking\ force\ at\ each\ wheel\ equally\ divided\ = 2893.125\ N$$